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The Remanent Magnetism of Some Lavas in the Deccan Traps†

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Abstract

The magnetizations of over 400 basalt specimens from the Deccan Traps have been found to be aligned in general along a NNW–SSE to NW–SE axis dipping towards the south. The results are consistent with the supposition that India has drifted northwards through over 50° of latitude and rotated 25° anticlockwise within the last 70 million years. In all three localities, the poles corresponding to the upper rocks are nearer the present geographical poles than those from the lower flows. The discrepancy may represent a northward drift of India during the formation of the Deccan Traps, but there are other possible explanations and further data are necessary to elucidate the problem.

§ 1. Introduction

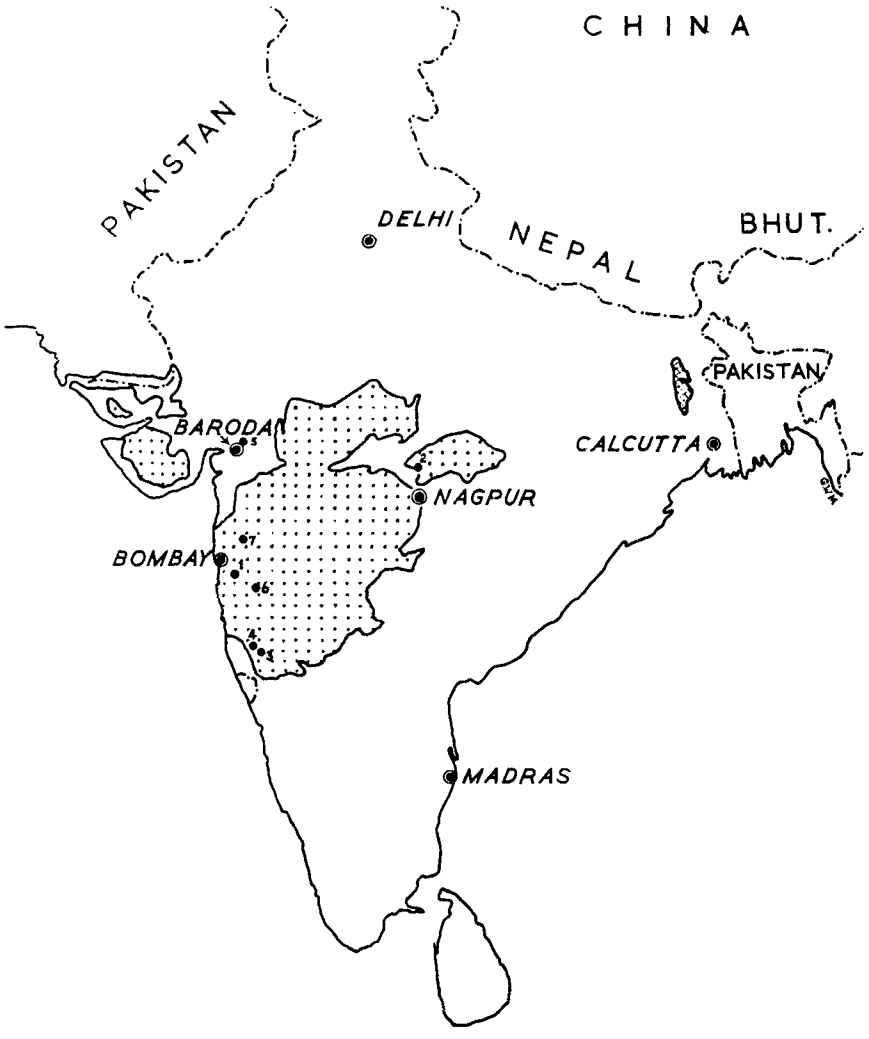
An earlier survey of the Deccan Traps (Clegg *et al.* 1956) was undertaken by the Physics Department of Imperial College in the winter of 1954–55 to initiate a project of palaeomagnetic research in India. Some 450 basalt specimens from two areas, Khandala and Linga (fig. 1, (1) and (2)), were examined at that time and shown to possess on the whole consistent remanent magnetizations whose direction is almost perpendicular to that of the present earth's field. The mean declination and dip for the two localities were N 155° E and 53° downwards respectively. This result is in good agreement with that found independently by Irving (1956), and is most plausibly interpreted on the assumption that at the time when the rocks were laid down, during the late Cretaceous or early Eocene periods, India occupied a position considerably to the south of the equator, and that it has since drifted northwards and rotated in an anticlockwise direction. The ancient positions of the North and South poles corresponding to this direction of magnetization differed strikingly from those obtained by other workers for British, North American and Icelandic rocks of the same period, and this suggested that at least part of the postulated movement of India may have been in relation to other continents.

The present investigation has been carried out with the aim of extending the previous work to other areas in the Deccan Traps. This is of considerable importance, since the formation covers a large area in Western

† Communicated by Dr. J. A. Clegg.

India (about 200 000 square miles), and attains a thickness of several thousand feet in some places, and it is highly probable that rocks from different areas may differ considerably in age. Moreover, by extending the survey, it was possible to compare magnetic results obtained from rocks of various petrological types, for in a number of localities in the Deccan Traps acid volcanic rocks occur in addition to the much more prevalent plateau basalts. One of the best known of such localities is

Fig. 1



Palaeomagnetic Surveys (Deccan Traps).

(1) Khandala area.
(2) Linga area.
(3) Nipani area.
(4) Amba area.
(5) Pavagadh area.
(6) Kambatki area.
(7) Igatpuri Ghats.

Mt. Pavagadh (fig. 1 (5)), a prominent hill northeast of the city of Baroda in the north of Bombay Province where basic as well as acid lavas were collected. In the south of Bombay Province, 400 miles from Mt. Pavagadh, two other localities were sampled in detail : the Nipani area (3), near the small town of that name on the Kolhapur–Belgaum highway, and the Amba area (4), constituting a hill of the Western Ghats about 50 miles northwest of Nipani on the Kolhapur–Ratnagiri highway. Apart from the three major surveys, two small test collections involving a small number of samples were made from two hills, the Kambatki (6) and Igatpuri (7) Ghats, situated between the two southern areas and Mt. Pavagadh. A preliminary and less complete account of measurements on rocks from Nipani, Amba, and Mt. Pavagadh has been given previously by Clegg *et al.* (1957).

§ 2. Collection and Measurement

Between October 1955 and November 1956, we undertook a number of field trips to the localities concerned and collected altogether 157 blocks of rock. At the collecting sites, these were marked with reference to the horizontal plane and to magnetic north, and the majority were taken to Bombay, where a palaeomagnetic laboratory suitable for the investigation of weakly magnetized geological specimens had been set up at the Tata Institute. Some of the rocks were shipped to Imperial College, London, for study. In each laboratory cylinders were cut from the blocks and the direction and intensity of their remanent magnetization were measured by means of a simplified adaptation of the astatic magnetometer described by Blackett (1952). In this, the standard procedure used by Clegg *et al.* (1954) was followed. In the present paper, the word ' specimen ' will be used to denote each cylinder cut from a block, and the word ' sample ' to denote a block. Altogether 480 specimens were measured, or an average of roughly three specimens to a sample. The results for the three areas are described separately below.

§ 3. The Nipani Area

The Deccan Trap flows in this region are exposed mainly on a number of low hills and in quarries. The present collection was made from two series of outcrops of basaltic lava, one occurring on the prominent escarpment lying three miles south of Nipani, and the other in two quarries on the plain ten miles north of Nipani. Samples were taken on the escarpment from five sites representing at least three distinct flows and a vertical column of nearly 200 feet. The single flow exposed in the quarries was 650 feet below the highest flow sampled on the escarpment, but since the two localities were about 13 miles apart, a regional northward tilt of the flows by about $\frac{1}{2}^\circ$ would suffice to place the quarry flow above the escarpment flows on the stratigraphic column. Despite the remarkable horizontality of the flows characteristic of most of the Deccan Traps (Wadia

Table 1. Summary of Investigations

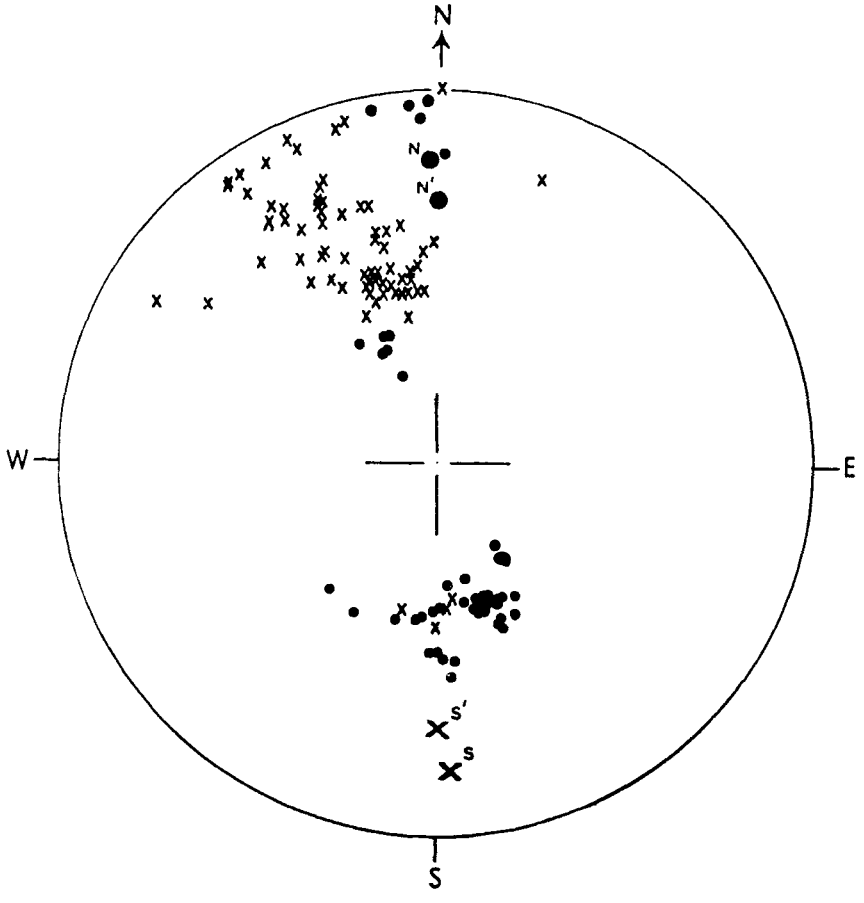
1 Area and section	2 Mean latitude (North)	3 Mean longitude (East)	4 Elevations (above mean sea level) (ft.)	5 Total No. of sites visited	6 Total No. of specimens measured	7 Mean intensity of magnetization ($\times 10^{-3}$ c.g.s.)	8 Mean declination (East of true North)	9 Mean dip[1] (positive downwards)	10 Radius of 50% circle of confidence[2]
Nipani, *Lower*	16° 32′	74° 19′	1850	2	44[3]	1·87	168°	+60°	1·9°
,, *Upper*	16° 21′	74° 24′	2325–2500	5	74[3]	1·96	338°	−32°	2·1°
Amba, *Lower*	17° 00′	73° 46′	500–1800	5	109[4] (17)[5]	3·49 (1·81)[5]	144° (159°)[5]	+60° (−75°)[5]	2·0° (4·5°)[5]
,, *Upper*	16° 59′	73° 47′	2050–2075	3	54	3·13	335°	−26°	3·3°
Pavagadh *Lower* (basic)[6]	22° 28′	71° 34′	1125–2425	8	69	2·22	351°	−16°	3·5°
,, (acid tuffs)	22° 28′	71° 33′	2490–2600	2	15	0·29	355°	+17°	3·4°
,, (rhyolites)[7]	22° 28′	71° 33′	2625–2725	2	69	0·59			

[1]Measured relative to a horizontal plane.
[2]Fisher's method (1953).
[3]Four specimens had inverted dips and are here omitted.
[4]Specimens with downward dips only.
[5]Specimens with anomalous upward dips.
[6]Chiefly basalt.
[7]Much of this group was randomly magnetized, so that a significant mean direction could not be computed.

1953), such a possibility cannot be completely ruled out. However, for the purpose of this paper, the quarries will be referred to as ' lower Nipani ', and the escarpment as ' upper Nipani '.

The results are shown in table 1 and the directions of magnetization of individual specimens are plotted in fig. 2. Specimens cut from one sample in each of the lower and upper sections showed scattered directions of magnetization, and these have been omitted. It is evident from fig. 2

Fig. 2



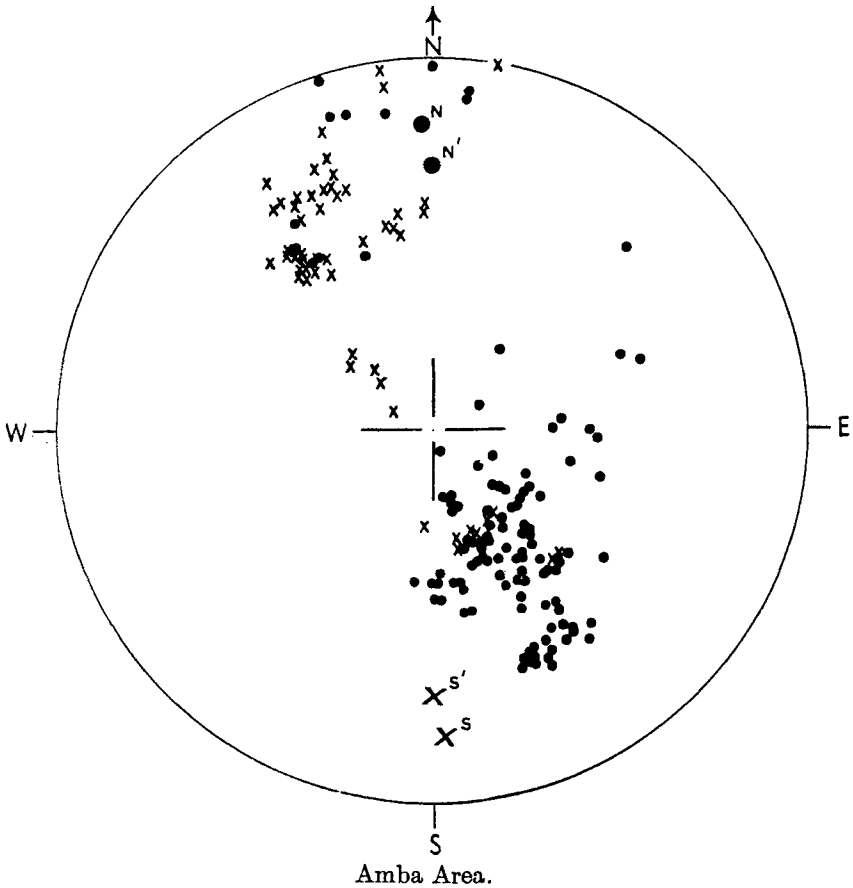
Nipani Area.
All Flows.
Directions of North-seeking Polarizations :
X Up ● Down
Directions corresponding to :
Present Poles :
● N X S
Dipoles :
● N1 X S1

that the magnetic directions are grouped into two main classes, the lower Nipani specimens generally having SSE declinations with rather steep downward dips, while most of the upper Nipani specimens have NNW declinations with shallower upward dips.

§ 4. The Amba Area

A vertical sequence of at least 1600 feet of basaltic lava flows is partly exposed on the steep side of the Amba Ghat, which lies on the western edge of the Deccan plateau just above the coastal plain. In the topography of the Western Ghats this hill occupies a place corresponding to the

Fig. 3



Amba Area.

Directions of North-seeking Polarizations :

X Up ● Down

Directions corresponding to :

Present Poles :

● N X S

Dipoles :—

● N1 X S1

Bhor Ghat at Khandala, 130 miles to the north, mentioned previously by Clegg *et al.* (1956). Eight sites representing five or more distinct flows were sampled and again the polarizations fell into two major groups (fig. 3 and table 1), with most of the directions aligned approximately along a NNW–SSE axis dipping downwards towards the south, similar to the preferential direction at Nipani. A sub-division into ' upper ' and ' lower ' Amba areas was made to distinguish the three highest sites, which were magnetized towards the NNW with an upward dip, from the five lower sites where the great majority of the samples had polarizations directed towards the SSE, but with rather steeper downward dips. As found in the Nipani area, a few of the lower Amba specimens yielded consistently odd results, with generally SSE declinations as expected, but also steep upward, instead of downward dips. Since 17 specimens from two samples and two different flows were here involved, values for their mean direction have been listed separately in table 1.

§ 5. The Pavagadh Area

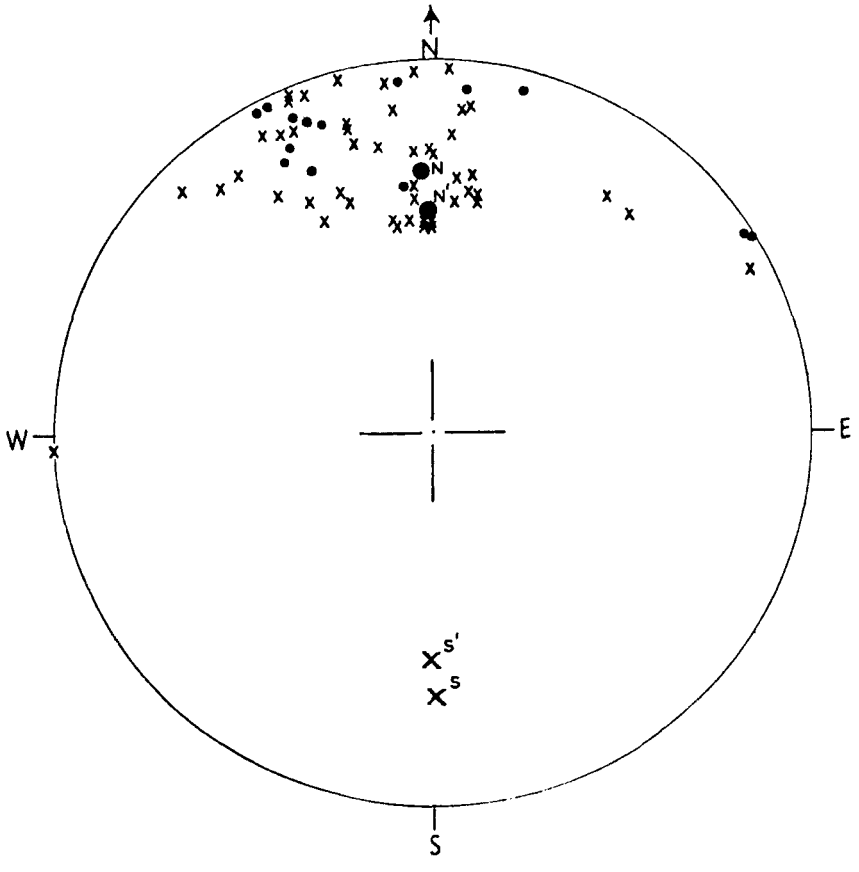
Mount Pavagadh is separated from the main body of the Deccan Traps by a distance of about 30 miles. It rises abruptly from the plains of Gajerat to a height of 2811 feet above sea level. The main mass of the hill, up to a height of about 2400 feet, consists of numerous nearly horizontally bedded flows of porphorytic as well as non-porphorytic basalt and olivine dolerite, together with some basic tuffs. Above this level the extensive Mauliya plateau is covered with green rhyolitic tuffs, and is capped with a large mass of rhyolite which rises steeply from the plateau to the summit. These acid rocks were first identified by Fermor (1906) who suggested that they might be interbedded with basic lavas, but Mathur and Dubey (1934) have since shown them to be younger than the basic lavas, and it is now believed that both the rhyolitic tuffs and the overlying rhyolites belong to an acid cycle of igneous activity which may have taken place considerably later than that which produced the basic rocks below. In the present account (see table 1) the lower basic rocks will be referred to as the ' Lower Pavagadh ' and the upper acid tuffs and rhyolites will be termed ' Upper Pavagadh '.

Eight flows of largely basic composition were investigated in the lower Pavagadh area. The magnetic directions (fig. 4) were predominantly N to NNW declinations with shallow upward dips. Fifteen specimens from eight samples of acid tuffs of the upper Pavagadh area were also examined. Their magnetizations were found to be directed nearly towards the north, as in the basic rocks, and the dips were again shallow, but slightly downwards. The data for the tuffs are plotted as open circles in fig. 5. The mean direction of magnetization is 5° west of north with shallow downward dip.

Directions for 69 specimens cut from 14 samples of rhyolite are also plotted on the same diagram. The results show too much scatter to permit the computation of a significant mean direction of magnetization. Different

specimens of rhyolite, even from samples collected at points not far apart, gave in some cases results which completely disagreed with one another, and they reacted quite differently to simple tests for their stability. The close spacing of the cluster of dots on the lower left side of fig. 5 would alone suggest that the 16 specimens concerned have stable magnetizations, particularly in view of the large angles which these make with the geomagnetic field. To test this, a number of the specimens were subjected

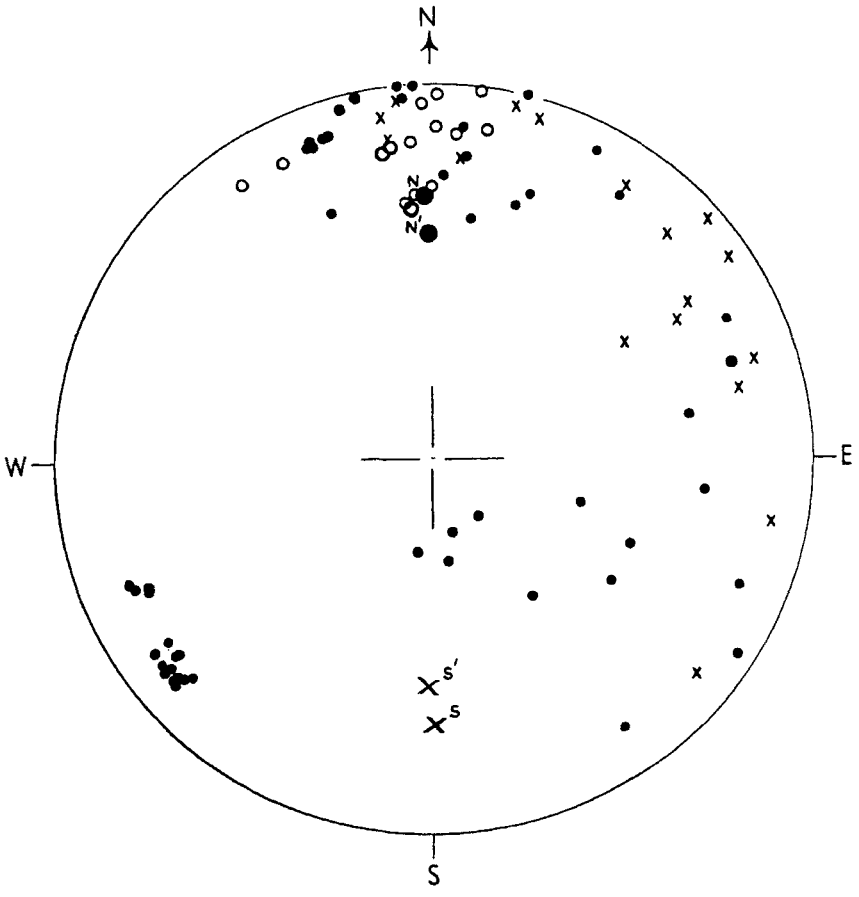
Fig. 4



Pavagadh Area.
Flows 1–8.
Directions of North-seeking Polarizations :
X Up ● Down
Directions corresponding to :
Present Poles :
● N X S
Dipoles :
● N1 X S1

to a.c. demagnetizing fields of 200 oersteds r.m.s.; this had no effect on their remanent polarizations. At the other extreme, when some of the specimens from samples showing much scatter were placed for periods varying from a few hours to a few weeks with their magnetization axes at

Fig. 5



Pavagadh Area.
All acid rocks (Tuff and Rhyolite).
Directions of North-seeking Polarizations :
X Up (Rhyolites).
● Down (Rhyolites)
○ Down (Tuffs)

Directions corresponding to :
Present Poles :
$\bullet^{N}$ X^{S}
Dipoles :
$\bullet^{N^1}$ X^{S^1}

right angles to the earth's field, and re-measured, this axis was usually found to have been pulled towards the magnetic meridian, in some cases through a complete right angle.

§ 6. Test Collections

Four samples each from two flows of the Kambatki Ghat south of Poona, and from four flows of the Igatpuri Ghat northeast of Bombay, were examined. The lower Kambatki flow gave a mean declination of 176° and a mean dip of +60°, whilst the corresponding values for the three upper Igatpuri flows were 161° and +51°. These results are similar to those obtained for the Khandala, Linga, lower Nipani and lower Amba areas, but not enough samples have been examined to justify any further conclusions. The upper flow at the Kambatki Ghat and the lowest flow at the Igatpuri Ghat had respectively NE and NW declinations with steep downward dips, perhaps due to partial instability.

§ 7. Discussion of Results

The complete results for the Nipani, Amba, and Pavagadh areas are summarized in table 1. The general direction of magnetization is along a NW–SE to NNW–SSE axis dipping towards the south. This is in broad agreement with the earlier results of Irving (1956) and Clegg *et al.* (1956), and supports the supposition that India has moved northwards across the Equator relative to the earth's geographical axis, and rotated in an anticlockwise direction, since the rocks were laid down. In the Nipani and Amba areas the lower flows have reverse (southeasterly) magnetizations with rather steep downward dips, while the uppermost flows are normally magnetized and the dips are significantly less. In the Pavagadh area there is also a significant difference between the lower basic rocks and the acid tuffs above, both having normal polarizations and shallow dips, but whereas the dips of the lower flows are upwards, those of the overlying tuffs are downwards. No significant mean direction could be computed for the rhyolities overlying the tuffs.

7.1. *The Directions of the Magnetic Axes and Ancient Pole Positions*

Since the period of time over which the rocks from each individual site were extruded is likely to have been large in comparison with the period of secular variation, it is reasonable to suppose that the magnetic directions shown in table 1 correspond to those of a central magnetic dipole coaxial with the earth's rotational axis. In column 4 of table 2 the values of the ancient latitude λ of the various sites have been computed on this assumption, from the mean dip θ, using the formula

$$\tan \lambda = \tfrac{1}{2} \tan \theta.$$

Table 2

1 Area and section	2 Mean declination (East of true North)	3 Mean magnetic dip	4 Ancient latitude of site	5 Ancient pole position		6 Ancient latitude of Bombay	7 Ancient magnetic dip at Bombay	8 Sense of magnetization
				Latitude	Longitude			
Linga *complete*	164°	+48° down	29° S	37° N	83° W	31° S	+50° down	Reversed
Khandala *complete*	147°	+50° ,,	39° S	25° N	79° W	39° S	+58° ,,	,,
Nipani *lower*	168°	+60° ,,	41° S	35° N	95° W	39° S	+58° ,,	,,
Amba *lower*	144°	+60° ,,	41° S	27° N	76° W	39° S	+58° ,,	,,
Nipani *upper*	338°	−32° up	17° S	52° N	70° W	14° S	−27° up	Normal
Amba *upper*	335°	−26° ,,	14° S	52° N	64° W	12° S	−23° ,,	,,
Pavagadh *lower*†	351°	−16° ,,	8° S	58° N	91° W	12° S	−23° ,,	,,
Pavagadh *upper*‡	335°	+17° down	9° N	75° N	89° W	5° N	+10° down	,,

† Mainly basalt. ‡ Acid tuffs only.

The ' reversely ' and ' normally ' magnetized series of flows have been grouped together for comparison. Column 5 of table 2 shows the ancient positions of the North geographical pole calculated from the values of λ, and from the mean declinations. These pole positions are compared with those obtained previously by Clegg *et al.* (1956). The few anomalous results from the various sites (e.g. the 17 specimens from the lower Amba area showing upward dips) have been omitted from this table.

The discrepancies between the values of λ shown in column 4 arise partially because the various sites are separated by distances of several hundreds of miles, and a better comparison is obtained by referring all the results to a single locality. For this purpose Bombay (present latitude 19° N) has been chosen, and columns 6 and 7 of table 2 show the ancient latitude and the ancient axial dipole field dip at Bombay, corresponding to the pole positions in column 5.

7.2. *Possible Implications of the Results*

From the summarized results set out in table 2 it is apparent that the rocks from the various localities visited fall into three main groups, viz.:

(*a*) Those of the lower Nipani and lower Amba sections, for which the magnetic directions and ancient pole positions correspond reasonably well to those obtained previously for the Linga and Khandala areas by Clegg *et al.* (1956).

(*b*) Those of the upper Nipani and upper Amba sections together with the basic rocks of lower Pavagadh, which are normally magnetized and have markedly shallower upward dips, and for which the corresponding ancient pole positions lie considerably closer to the present geographical pole.

(*c*) The upper rhyolitic tuffs of upper Pavagadh, which are again normally magnetized, but which have a shallow downward dip corresponding to a low northerly latitude.

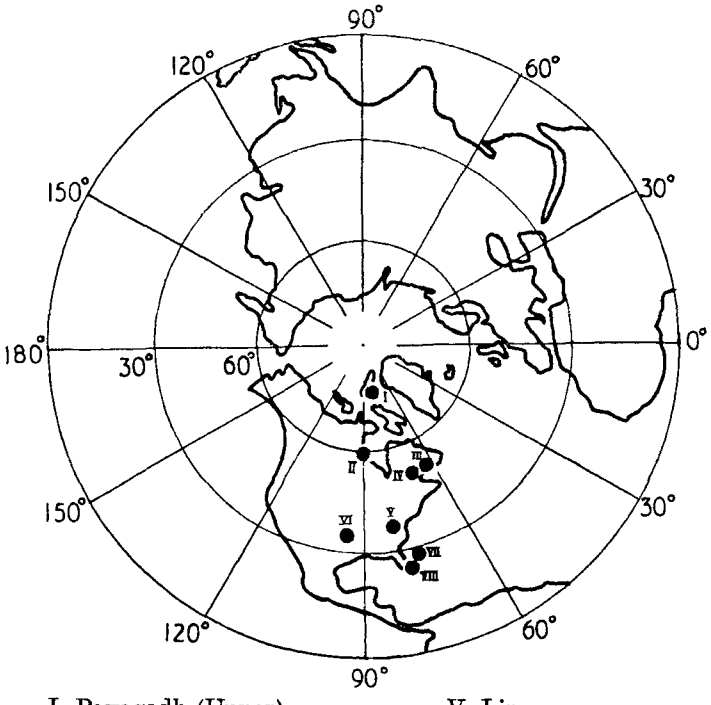
The mean directions of magnetization for these three groups, obtained from table 2 (directions are referred to Bombay) are

	Mean declination	Mean dip
(*a*)	N 156° E	+56° down
(*b*)	N 341° E	−24° up
(*c*)	N 355° E	+10° down

One obvious possible explanation of these facts is that the three groups of rocks were laid down during different geological epochs, between each of which India moved northward relative to the earth's geographical axis. It may also be that the earth's magnetic field underwent a reversal between the first and second of the epochs. The geological evidence provides some support for this view. Thus in the Amba Area there is a section 200–300 feet thick between the uppermost ' reversed ' and lower ' normal ' flows in which no exposures were found, so that the upper and

lower sections probably differ considerably from one another in age. Again in the Nipani area a horizontal gap of 13 miles, containing four outcrops, separates the two localities, so that a strong possibility exists that the missing section represents a relatively long time interval, although it is not definitely established that the ' upper ' Nipani rocks are indeed younger than those of the ' lower ' Nipani area. The unusual sequence on Mt. Pavagadh is of particular interest. The lower basic rocks may well belong to the uppermost flows of the Deccan Traps, and it is probable that a considerable time elapsed between the last basalt outpouring and the beginning of the acid cycle responsible for the overlying rhyolitic

Fig. 6



I. Pavagadh (Upper)
II. Pavagadh (Lower)
III. Amba (Upper)
IV. Nipani (Upper)
V. Linga
VI. Nipani (Lower)
VII. Amba (Lower)
VIII. Khandala

tuffs. Mathur and Dubey (1934) quote results of radioactive age determinations of basalt and rhyolite specimens from Mt. Pavagadh, made by the helium method. Although helium dating seldom yields precise results they estimate from their data that the age of the basalt is early Eocene, in conformity with the remainder of the Deccan Traps, while that of the rhyolite is Miocene. If we assume the position of India at the end of the Cretaceous period to be that originally postulated by Clegg *et al.* (1956) and by Irving (1956), and the drift to have proceeded at a more

or less constant rate up to the present time, the area under investigation should have crossed the equator during the early Miocene. The observed mean dip of the tuffs would tend to support this supposition.

On the other hand, it is by no means certain, on the basis of the evidence so far available, that this variation of magnetic dip with depth can be taken as positive evidence of a continuous northward movement of India. One major difficulty in the interpretation of the results arises from the uncertainty in the relative dating of the basic rocks from the various sites. For the total depth of the Deccan Traps, particularly towards their eastern edge, is known to exceed 10 000 feet in places (Wadia 1953), and as it was only possible to sample a small fraction of this range at any of the sites visited, it seems improbable on primae faciae considerations that the rocks of either of the two categories (*a*) and (*b*) above are from corresponding parts of the stratigraphical column.

There are obviously other possible physical causes which may have given rise to the difference in inclination between the categories (*a*), (*b*) and (*c*). It may, for instance, be due to some physical phenomenon associated with different depths of burial, such as the magnetostriction effect suggested by Graham (1957). Alternatively it may result from partial magnetic instability, and the consequent pulling of the magnetic directions towards that of the present earth's field. For rocks which are magnetized normally this latter effect would tend to reduce the magnetic dip, while for those magnetized reversely it would tend to increase the dip. In this connection it is worth noting that the rocks of category (*a*) which have the steepest dips, are reversely magnetized, while those of categories (*b*) and (*c*), with shallower dips, are all normal.

Work is continuing in an attempt to find exposures of intermediate sections in the areas sampled so far, and the survey is also being extended to include a detailed study of vertical sequences in other parts of the Deccan Traps.

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